

Dynamic Multi-Spectral Adversarial Camouflage Pattern Generation for Adaptive Object
Concealment in Visible and Infrared Domains

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ABSTRACT

The recent times have seen astronomical advancement in the deep-learning based object detection systems, especially in the visible and the infrared (IR) spectra. This has significantly reduced the effectiveness of conventional camouflage strategies. Modern day surveillance systems employ multi-spectral sensing, enabling robust detectability under varying environmental and illumination conditions. Although adversarial machine learning systems have given birth to multiple methods and frameworks that generate carefully optimized patterns which can successfully mislead even the best state-of-the-art object detectors, existing approaches primarily focus on static perturbations within a single spectral domain and rarely consider dynamic motion or cross-spectral consistency.

This research proposes MS-DACS (Multi-Spectral Dynamic Adversarial Camouflage System), a motion-aware framework for generating adaptive camouflage patterns designed to attack both the visible and the infrared domains. MS-DACS models object dynamics using temporal bounding box displacement and scale variation, which are used to condition a generative network for frame-wise camouflage pattern synthesis. A multi-objective optimization strategy integrates detection suppression, spectral consistency and temporal smoothness constraints to ensure realistic and stable pattern generation across consecutive frames

The proposed system will be evaluated using the KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015), with performance assessed through detection confidence reduction, attack success rate, and robustness under dynamic motion scenarios. By integrating motion modelling with multi-spectral adversarial optimization, MS-DACS advances intelligent camouflage design beyond static perturbation methods toward adaptive, sensor-aware concealment mechanisms.

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LIST OF ABBREVIATIONS

MS-DACS – Multi-Spectral Dynamic Adversarial Camouflage System

CNN – Convolutional Neural Network

R-CNN – Region-based Convolutional Neural Network

IR – Infrared

RGB – Red-Green-Blue

RADAR – Radio Detection and Ranging

YOLO – You Only Look Once

TPM – Triplanar Mapping

GAN – Generative Adversarial Network

cGAN – Conditional Generative Adversarial Network

mAP – Mean Average Precision

LPIPS - Learned Perceptual Image Patch Similarity

ASR – Attack Success Rate

CPU – Central Processing Unit

GPU – Graphics Processing Unit

RAM – Random Access Memory

VRAM – Video Random Access Memory

SSD – Solid State Drive

RTX - Ray Tracing Texel eXtreme

LTS – Long-Term Support

1. INTRODUCTION

Camouflage has always been considered one of the most effective survival strategies in nature, allowing organisms an opportunity to either avoid predators, or approach the prey undetected. Early studies on biological camouflage strategies, as documented by Thayer Gerald Handerson and Thayer Abbott Handerson (1909) describe in great detail several mechanisms including countershading, disruptive colouration and background matching, all of which successfully reduce the visual detectability of an organism. Subsequent work done by Cott (1940) further expanded on these concepts by analysing additional deceptive strategies, such as masquerade, mimicry and warning colouration. Natural principles such as these have significantly influenced the design of artificial camouflage strategies employed in the military and surveillance domain.

However, the challenge to remain hidden has become increasingly complex with the rapid development of modern sensing technologies. Developments in deep learning have led to highly sophisticated and accurate object detection systems capable of identifying targets across multiple modalities, including visible (RGB) and thermal (IR) spectra. These systems are widely deployed in surveillance infrastructures, such as autonomous driving and defence technologies. This has rendered the traditional camouflage techniques which primarily rely on static colour blending, obsolete against such advanced machine perception systems, that analyse high dimensional visual features across multiple spectral domains.

To address these challenges, researchers have explored the use of adversarial machine learning to design camouflage patterns, capable of misleading modern deep neural network-based detectors. Early works on adversarial patch proved that carefully optimized perturbations could cause object detectors to misclassify or even fail to detect objects entirely. Subsequently, researches expanded into the concepts of adversarial camouflage textures that would integrate natural looking perturbations onto objects. Recent approaches have even investigated physically realizable camouflage, capable of reducing detection performance under varying viewpoints and environmental conditions.

Despite such advancements, majority of modern adversarial camouflage techniques rely on static patterns that remain fixed once applied. Such patterns, may exhibit robustness against object transformations such as rotation or scaling, they fall short when faced against dynamic environmental changes. Furthermore, it is to be noted that existing studies focus on a single, primarily the visible spectrum, whereas most modern detection systems operate across multiple

spectral modalities. These limitations greatly restrict the effectiveness of current camouflage strategies in realistic conditions where dynamic environments and cross-modalities are real.

Faced with such challenges, the research proposes MS-DACS (Multi-Spectral Dynamic Adversarial Camouflage System), a novel motion-aware framework, designed to generate camouflage patterns objects in visible and infrared domains. The proposed approach models motion from the bounding box’s temporal variations and extracts background context feature to condition a generative camouflage network. By integrating motion modelling, background feature and adversarial optimization, the framework generates frame-wise camouflage patterns, capable of reducing detection performance across visible and infrared domains, while maintaining visual consistency.

2. LITERATURE REVIEW

Research on camouflage has been going on for decades, with researchers trying to achieve something close to invisibility. Hwang et al. (2015) were one of the first to introduce in their paper the ability to detect pedestrians in visible and infrared spectrums. Ever since, researchers have strived to introduce camouflage systems to fool such detection algorithms.

One of the first such success came from Brown et al. (2018) where they would replace a part of an image(object) with their patch(sticker). The results were fed to a Whitebox-Single model, Whitebox-Ensemble and Blackbox, which successfully broke the classifiers.

While Brown et al. (2018) only attacked the classifiers, causing a wrong class to be inferenced, Liu et al. (2018) furthered the work by not only ensuring a wrong image-level classification, they also attacked the bounding box regression to disable their prediction. In their paper, the default patch was a 40 – by – 40 square, attached to the top-left corner of the image. YOLO and Faster R-CNN failed to recognize any objects when fed with the resultant image. Liu et al. (2018) also proved the transferability of their attack where the YOLO-trained patch had successfully disabled Faster R-CNN.

Chen Shang-Tseand Cornelius (2019) introduced *ShapeShifter*, a robust targeted attack on image-based object detectors, by adapting the **Expectation over Transformation** technique originally proposed for image classification. In their paper, Chen Shang-Tseand Cornelius (2019) generated adversarially perturbed stop signs that were mis-detected by Faster R-CNN. Thys et al. (2019) then in their paper, generated adversarial patches that were used to attack

person detection. They used the Inria dataset, which were targeted more towards full-body pedestrians and for the detector, Thys et al. (2019) used the YOLOv2 object detector.

Duan et al. (2020) took a very different approach in their paper, where they introduced *Adversarial Camouflage (AdvCam)* which integrated natural styles onto the adversarial examples. This provided a unique approach where while Brown et al. (2018) and their successors generated a patch that was easily identified by human observers, patches by Duan et al. (2020) appeared legitimate to humans, yet stayed “hidden” to object detectors. Duan et al. (2020) achieved this by developing a mechanism such that the attacker can flexibly decide on the attack region and/or the attack style which then generates the final adversarial image.

Zhu et al. (2021) in their paper focused on thermal (IR) attacks instead of visible spectrum like their predecessors. The methodology used by Zhu et al. (2021) in their paper was to introduce a board strewn with small light bulbs. They attacked state-of-the-art pedestrian detector based on YOLOv3, which caused the average precision (AP) to drop by 64.12% in the digital world and by 34.48% in the real world.

With adversarial camouflage brewing up a storm in the research domain, Suryanto et al. (2023) came up with the Adversarial Camouflage for Transferable and Intensive Vehicle Evasion (ACTIVE) framework, focused on vehicles. In their paper, Suryanto et al. (2023) used Tripalanar Mapping (TPM) to apply the textures on the objects. They have also implemented Stealth Loss, Smooth Loss and Camouflage Loss for proper evaluation.

Talas et al. (2020) introduced a novel method of generating camouflage patterns using Generative Adversarial network (GAN), where patterns generated after more iterations were harder to detect with the generator reaching optimal solution at around 5000 iterations. In 2024, came a paper named “CamoDiff” by Nguyen et al. (2024) where they proposed a few-shot diffusion-based method for camouflage pattern generation. The method was indeed revolutionary where the patterns generated had an average mAP as low as around 8.2.

Wei et al. (2024) proposed a transferable adversarial attack method in their paper. They designed a classification probability loss based on inter-class probability distribution and an IoU loss function based on the predicted bounding box to train their pattern generator. Wei et al. (2024) in the paper claimed that when the generated pattern was applied to the hood of the cars, there was a staggering 91.2% and 90.7% reduction in car detection respectively against YOLOv3 detector.

Even though camouflage is a technique used for stealth, having a patch/pattern applied to a static object often loses its meaning when the object needs to be moving. The Dynamic Adversarial Patch by Hoory et al. (2020) introduced the dynamic nature into the adversarial patch where the adversarial patch would be applied to the object on pre-determined locations. The dynamic attack was implemented by switching the appropriate patches on the object according to the camera's position, i.e. the detector's location. When tested on the YOLOv2, the detector was misled in 90% of the video frames. Hoory et al. (2020) also claimed it to be highly transferable among other car models with where the detector was successfully misled about 71% of the time.

Zhu et al. (2025) further improved Hoory et al. (2020)'s work in their paper where they introduced an approach named CamoEnv, where they used the nuScenes dataset and Carla simulator and three car models from the IM3D dataset. They prepared two sets of tasks, one was to generate a pattern for each scene/frame and the other was to generate a universal pattern, emphasizing generalizability. Zhu et al. (2025) attacked 14 different detector models and CamoEnv had better ASR(%) score and environmental LPIPS(%) score across the board.

Recent years have witnessed groundbreaking advancements in adversarial camouflage research across the globe, spanning both the visible and infrared (IR) spectra. Most existing studies, however, focus on scenarios where either the object or the detector remains static. With the rapid evolution of modern technology and its growing impact on defence and security, there is now a pressing need for methods capable of dynamically challenging detectors across both spectral domains.

3. AIM

To design and evaluate MS-DACS, a motion-aware multi-spectral adversarial camouflage system capable of generating camouflage patterns that reduce object detectability in both visible and infrared domains.

4.OBJECTIVES

1. To model object dynamics using temporal bounding box displacement and variation in scale from sequential frames.

2. To develop a generative framework that generates frame-wise adaptive camouflage patterns based on object’s motion and background context.
3. To design a novel cross-spectral optimization strategy that ensures consistency between visible as well as infrared camouflage patterns.
4. To evaluate the effectiveness of MS-DACS using KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015), measuring detection confidence reduction, attack success rate, and robustness under dynamic motion scenarios.
5. To compare the proposed dynamic approaches against existing methods and frameworks.

5. SIGNIFICANCE

The evolution of deep learning-based object detection has reached unprecedented heights, operating in visible and infrared spectra, significantly enhancing surveillance capabilities. Conversely, this has also highlighted the shortcomings of traditional camouflage strategies, which are predominantly static and focused on a single spectrum. As multi-spectral surveillance becomes prevalent in today’s defence, security and automated monitoring systems, the need for a dynamic, adaptive camouflage strategy that account for cross spectral consistency is becoming nothing but necessary.

This research aims to introduce camouflage generation beyond static perturbations toward dynamic, motion-aware adaptation in the field of adversarial machine learning. The proposed MS-DACS introduces a novel perspective on intelligent camouflage design by integrating temporal modelling with multi-spectral optimization.

6. SCOPE

This research focuses on developing and evaluating a dynamic adversarial camouflage pattern generation framework that operates in visible and infrared domains. The research utilizes KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015) to model object motion and detection suppression evaluation performance under cross-spectral conditions.

The proposed framework generated frame-wise camouflage patterns based on temporal motion descriptions derived from bounding box displacement and scale variations. Evaluation will be

conducted against selected deep learning-based object detectors to measure attack success rate and reduction in detection confidence.

It is to be noted that the scope is explicitly limited to digital simulations and does not include physical fabrication, material designing, real-world deployment or hardware implementation. Additionally, the research focuses on pedestrian detection scenarios and does not extend to vehicles or aerial systems.

7. METHODOLOGY

The **Dynamic Adversarial Camouflage Generation** framework proposed in this research aims to produce frame-wise adaptive camouflage patterns for objects, both in the visible and infrared spectra.

Unlike the existing adversarial camouflage works done by researchers, which predominantly optimize a single static texture, the proposed framework generates context-aware camouflage patterns that change according to background appearances and object's motion over time. The framework leverages a deep generative model trained on environmental context to produce camouflage textures that:

1. Blend with the surrounding scene
2. Reduce detection confidence of object detection models

The system operates entirely in a simulated pipeline using the KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015), where synthetic motion is applied to generate frame sequences. For each frame, local background features around the objects are extracted and used as conditioning information for a camouflage generation network.

The generated camouflage is then applied to the object, and the modified frame is run through pretrained object detectors such as YOLO and Faster R-CNN. An adversarial optimization process updates the generator to maximize detection reduction while preserving realism.

Figure 8.1 shows the overall framework of MS-DACS, where we have paired visible and thermal images. These images undergo data preprocessing in the forms of transformations such as translation, rotation and scaling. Post data preprocessing, the paired images are run through our pre-trained object detectors to obtain the bounding box coordinates. The bounding box

coordinates are then expanded to get the background features which along with a random noise vector are fed to our adversarial camouflage pattern generator.

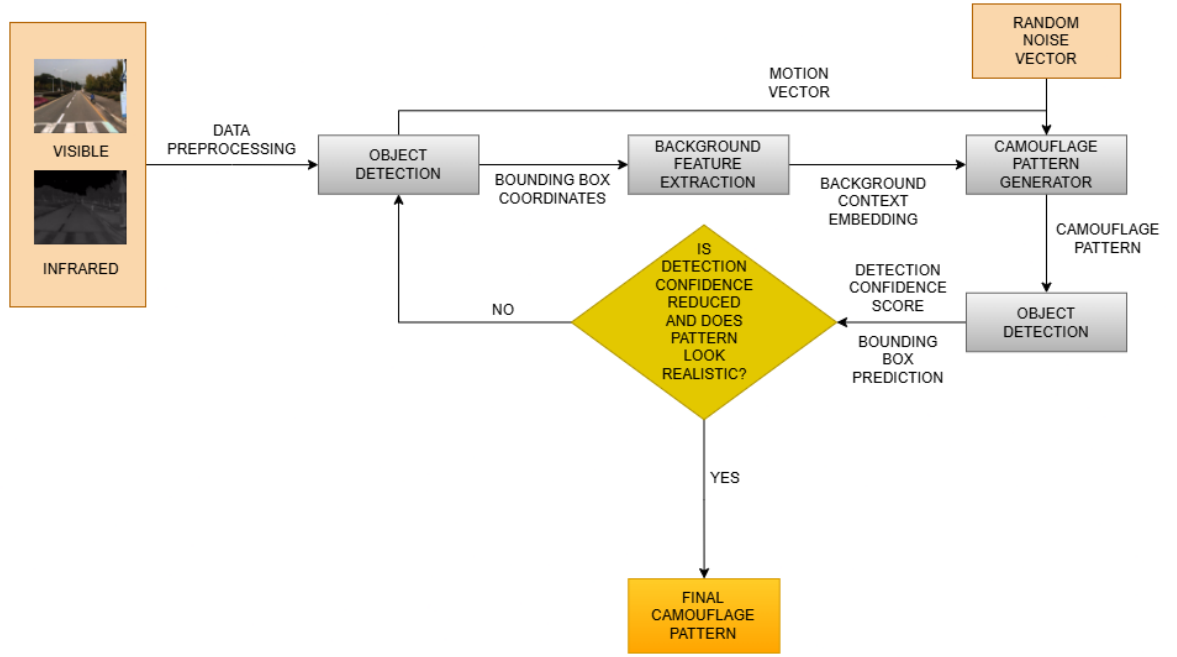


Figure 8.1: Overall framework of the research methodology

The generated camouflage pattern is again run through pre-trained detectors for evaluation. If the generated pattern fails in misleading the detector and looks realistic, the training continues till the results are satisfactory.

7.1. MULTISPECTRAL DATASET PREPARATION

The proposed system uses the KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015), which provides us with paired RGB and Infrared images, captured in real-world urban environments. These images allow the study of camouflage effectiveness across visible (RGB) which is used by standard surveillance, computer vision systems and humans, as well as thermal (IR) which is commonly used in nighttime surveillance and autonomous driving.

To simulate motion scenarios, synthetic frame-wise transformations are applied to pedestrian bounding boxes. The transformations that are going to be implemented are as follows:

Translation: The object will be slightly shifted in either horizontal or vertical direction to simulate motion.

Scaling: This transformation causes the object to be resized, by either enlarging or shrinking it by a certain factor.

Rotation: The objects will be turned or rotated around a fixed point (usually around the centre) by an angle, keeping the shape and size constant.

This process creates a pseudo-video sequence that allow the proposed framework to generate dynamic camouflage patterns over time.

The dataset preparation stage produces paired visible and IR frame sequences with transformed objects, ready for object detection and background context extraction.

7.2. OBJECT DETECTION AND TARGET REGION IDENTIFICATION

The pretrained object detectors are used on each frame to identify the target object region. The aim of this stage is to:

- Locate the object region where camouflage will be applied.
- Provide object detection confidence scores, which will later be used in adversarial optimization.

Detectors incorporated during training will be an ensemble of YOLO based object detectors, such as YOLOv8 by Varghese and M. (2024) and Faster R-CNN by Ren et al. (2016)

Utilizing multiple object detectors during training forces the framework to generalize for unseen detectors as well rather than focus on attacking one type of detector.

The output of this stage includes axis-aligned bounding box coordinates of the objects and detection confidence scores, which is a product of the probability that the object exists within the bounding box coordinates and the probability of the class identification.

7.3. BACKGROUND CONTEXT EXTRACTION

To generate realistic camouflage patterns, the model needs to first understand the visual context of the object's surroundings.

Once the target object is detected, the corresponding bounding box is utilized to define a local context window. This is achieved by expanding the bounding box by a fixed predefined scale to capture the immediate surrounding. To ensure the background features does not include the object itself, we will mask the pixels of the objects original bounding box. This masking process

isolates the immediate background region while preventing the object’s appearance from influencing the background representation. From this region, the system derives the following:

- Dominant colours
- Texture statistics
- Spatial patterns
- Thermal intensity distributions

These background features are encoded using a convolutional feature extractor, producing a context embedding vector, representing the local environment. This is then used as conditioning input to the camouflage pattern generator.

The goal of this step is to ensure that generated patterns visually and thermally blend with the surrounding scene.

7.4. CONTEXT CONDITIONED CAMOUFLAGE PATTERN GENERATION

The core component of the framework is a deep generative model that produces camouflage textures conditioned on environmental context.

The generator receives three major inputs, namely the background context embedding, random latent noise vector and motion vector. The random noise is a simple random numerical vector to introduce a source of randomness into the training, so that the patterns generated are stochastic instead of deterministic. The motion vector is the representation of the object’s motion in terms of vertical translation, horizontal translation and scale variation.

These inputs are used to synthesize a camouflage texture map, that:

- Matches the surrounding background appearance
- Contains adversarial perturbations that disrupts object detectors

For the generator, we are focusing on cGAN. The reason behind choosing cGAN over a diffusion based generative model or an encoder-decoder based generative model are cGANs are relatively lightweight. cGANs draw inference faster than a diffusion based generative model and produces more realistic textures than an encoder-decoder based generator.

The generated output is a camouflage texture that can be mapped onto object’s surface. Since the generator operates on a per-frame basis, the resulting pattern changes dynamically as the object moves through different backgrounds.

7.5. CAMOUFLAGE TEXTURE APPLICATION

Once the pattern is generated, it is mapped onto the object surface. The process includes texture scaling to match the object size, spatial alignment with bounding box, which we will be applying via masking, and blending with original object appearance. Blending is achieved by alpha blending, which is a weighted combination of original object pixels and camouflage texture.

To improve realism, smoothness and colour consistency may be applied during blending.

7.6. DETECTOR EVALUATION AND ADVERSARIAL FEEDBACK

The modified frame is evaluated using the pretrained object detectors. For each frame the detectors provide the following, namely Detection confidence scores, Bounding box predictions and Feature activations.

These are used to compute an adversarial loss, which measures how successful the camouflage pattern is at reducing detection. These losses further guide the training process of the camouflage pattern generator.

7.7. OPTIMIZATION AND MODEL TRAINING

The generator is trained using gradient-based optimization, where the adversarial loss is backpropagated through the pipeline. The training objective is to produce camouflage patterns that can simultaneously:

- Reduce detection accuracy
- Maintain visual realism
- Adapt to changing environmental context

Over multiple training iterations, the model learns to generate robust dynamic camouflage patterns, capable of degrading detector performance across various backgrounds and sensing modalities.

8. REQUIREMENT RESOURCES

We will need the following resources for development of the framework:

- COMPUTING HARDWARE: Personal laptop with the following configurations:
 - Processor: 12th Gen Intel® Core™ i5-12500H (16 CPUs), ~2.5GHz
 - Memory: 16384MB RAM
 - Operating System: Ubuntu 20.04 LTS (Focal Fossa)
 - Storage: 931GB SSD
 - GPU: NVIDIA GeForce RTX 3050 Laptop GPU
 - VRAM: 3964MB
- CLOUD SOLUTIONS: Google Colab for more GPU intensive training or tasks which may prove very harsh or time-consuming on personal system.
- SOFTWARE AND LIBRARIES: We will be using Python and its huge assortment of libraries such as NumPy, TensorFlow, Matplotlib etc.
- DATASET: For the dataset, we will be using the KAIST Multispectral Pedestrian Dataset by Hwang et al. (2015).

9. RESEARCH PLAN

Our research is divided into multiple milestones which we will be covering over the span of 10 weeks. The detailed plan is as follows:

- WEEK 1: For the first week, we are targeting further detailed literature review and starting with the data preparation. Objectives include in-depth analysis and identifying trends and gaps in recent papers, understanding the KAIST dataset and performing preliminary analysis (if required). *Milestone*: Finalize available relevant techniques and complete dataset understanding and analysis.
- WEEK 2: The second week will witness the object detection, background context extraction and motion simulation. We will be using YOLO based model for object detection, bounding box and background context extraction. For the motion simulation, we will be doing transformations in the form of translation, rotation and scaling. *Milestone*: Data pre-processing and preparation.

- WEEK 3: In the third week, we will target to build the Camouflage Generator architecture. We will primarily work with Conditional GAN for the generator and move into diffusion-based model if required or time permits. *Milestone:* Finalize generator architecture and start generating frame-wise camouflage patterns.
- WEEK 4-5: The fourth week will be focused on adversarial training, where the generated camouflage patterns are applied onto the objects in each frame and loss will be calculated. *Milestone:* Adversarial training against multiple detectors to significantly reduce detectability.
- WEEK 6: By the sixth week, the training will be completed and we will move onto the evaluation stage. We will use multiple detectors and compare against recent papers to verify the success of the research. *Milestone:* Evaluation and comparison against recent works and prepare appropriate charts and tables.
- WEEK 7: With the heavy-lifting done, we will move onto drafting the research report. We will be fine-tuning the necessary sections and prepare all figures, charts and tables according to our research journey. *Milestone:* prepare the full first draft of the research report.
- WEEK 8: The eighth week will be devoted to feedback and refinement process. Necessary revisions according to thesis supervisor's feedback will be carried out. *Milestone:* Final revised report with all feedbacks incorporated.
- WEEK 9: The ninth week will be kept as a buffer period to accommodate any unforeseen circumstances.
- WEEK 10: Submission of the research report post preparation of the final document.

The decided timeline is intensive and is designed so each milestone can be achieved within the allocated duration. Risks include potential delays in appropriate architecture selection and design, model training and lower-than-expected results. We have kept a buffer week to accommodate the above-mentioned risks or any other unforeseen events so the final submission deadline remains intact and untouched.

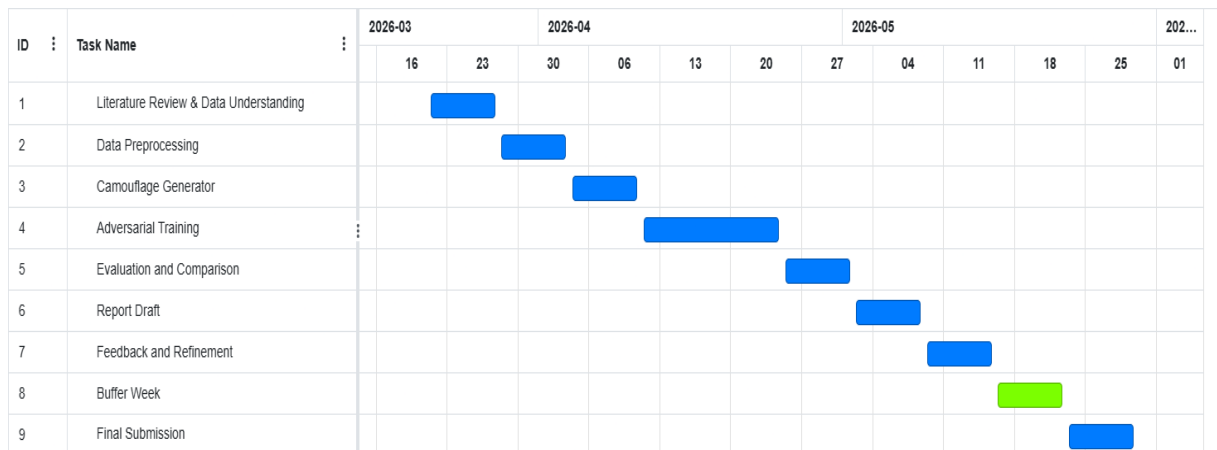


Figure 10.1: Grantt Chart

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